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IMPLANTABLE MEDICAL SYSTEM

Abstract

A medical system including a device head configured to be positioned in an atrium of a heart, an implantable medical device configured to be positioned within a vena cava of the heart, and a lead extending from the device head to the implantable medical device. A fixation element coupled to the device head is configured to engage tissues within the atrium. The device head includes an electrode configured to deliver therapy and/or sensing signals to tissues within the atrium using stimulation signals received from processing circuitry within the implantable medical device. The medical system may include a delivery catheter configured to allow delivery of the device head to the atrium.

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Background/Summary

[0001] This application is a continuation of U.S. patent application Ser. No. 17/453,044 (filed Nov. 1, 2021), which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/110,007 (filed Nov. 5, 2020), both of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] This disclosure is related to implantable medical systems.

BACKGROUND

[0003] Various types of implantable medical leads have been implanted for treating or monitoring one or more conditions of a patient. Such implantable medical leads may be adapted to allow medical devices to monitor or treat conditions or functions relating to heart, muscle, nerve, brain, stomach, endocrine organs or other organs and their related functions. Implantable medical leads include electrodes and/or other elements for physiological sensing and/or therapy delivery. Implantable medical leads allow the sensing/therapy elements to be positioned at one or more target locations for those functions, while the medical devices electrically coupled to those elements via the leads are at different locations.

[0004] Implantable medical leads, e.g., distal portions of elongated implantable medical leads, may be implanted at target locations selected to detect a physiological condition of the patient and/or deliver one or more therapies. For example, implantable medical leads may be delivered to locations within an atrium or ventricle to sense intrinsic cardiac signals and deliver pacing or antitachyarrhythmia shock therapy from a medical device coupled to the lead. In other examples, implantable medical leads may be tunneled to locations adjacent a spinal cord or other nerves for delivering pain therapy from a medical device coupled to the lead. Implantable medical leads may include anchoring components to secure a distal end of the lead at the target location.

[0005] Some cardiac pacemakers are sized to be completely implanted within one of the chambers of the heart, and may include electrodes integrated with or attached to the device housing rather than leads. Such cardiac pacemakers may include anchoring components to secure the pacemaker to cardiac tissue at a target location. Some cardiac pacemakers provide dual chamber functionality, by sensing and/or stimulating the activity of both atria and ventricles, or other multi-chamber functionality. A cardiac pacemaker may provide multi-chamber functionality via leads that extend to respective heart chambers, or multiple cardiac pacemakers may provide multi-chamber functionality by being implanted in respective chambers.

SUMMARY

[0006] In an example, a medical system comprises: an implantable medical device having a proximal end and a distal end, wherein the implantable medical device is configured to be positioned within a vena cava of a heart; a flexible lead, wherein a proximal end of the flexible lead is attached to the distal end of the implantable medical device, wherein the flexible lead is configured to extend from the vena cava to an atrium of a heart; a device head, wherein a first side of the device head is attached to a distal end of the flexible lead, wherein the device head is configured to be positioned within the atrium of the heart; a fixation element attached to the device head, wherein the fixation element is configured to engage tissue to substantially secure the device head within the atrium of the heart; and an electrode coupled to the device head, wherein the electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the atrium of the heart.

[0007] In an example, a system comprises: a medical assembly comprising: an implantable medical

device having a proximal end and a distal end, wherein the implantable medical device is configured to be positioned within a vena cava of a heart; a flexible lead, wherein a proximal end of the flexible lead is attached to the distal end of the implantable medical device, wherein the flexible lead is configured to extend from the vena cava to an atrium of a heart; a device head, wherein a first side of the device head is attached to a distal end of the flexible lead, wherein the device head is configured to be positioned within the atrium of the heart; a fixation element attached to the device head, wherein the fixation element is configured to engage tissue to substantially secure the device head within the atrium of the heart; and an electrode coupled to the device head, wherein the electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the atrium of the heart; and a delivery catheter defining a lumen and defining a lumen opening at a distal end of the delivery catheter, wherein: the lumen is configured to surround at least the implantable medical device, the flexible lead, the device head, and the fixation element, and the fixation element, the device head, the flexible lead, and the implantable medical device are configured to pass distally through the lumen opening.

[0008] In an example, a technique comprises: positioning a device head, an electrode attached to the device head, and a fixation element attached to the device head within an atrium of a heart by transmitting an axial force to the device head; causing a flexible lead attached to the device head and attached to an implantable medical device to extend from the atrium of the heart to a vena cava of the heart using the transmitted axial force; and positioning the implantable medical device within the vena cava of the heart using the transmitted axial force.

[0009] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a conceptual diagram illustrating an example medical system and delivery catheter.

[0011] FIG. 2 is a conceptual diagram illustrating the medical system of FIG. 1 translated through the delivery catheter of FIG. 1.

[0012] FIG. 3 is a conceptual diagram illustrating the medical system of FIG. 1 positioned with a heart.

[0013] FIG. 4 is a conceptual diagram of an example medical system including a device head, a lead, and an implantable medical device.

[0014] FIG. 5A is an illustration of a delivery catheter.

[0015] FIG. 5B is another illustration of the delivery catheter of FIG. 5A.

[0016] FIG. 6A is an illustration of a delivery catheter within a heart.

[0017] FIG. 6B is another illustration of the delivery catheter of FIG. 6A within the heart.

[0018] FIG. 7 is an illustration of a medical system and delivery catheter within a heart.

[0019] FIG. 8A is an illustration of a medical system implanted within a heart.

[0020] FIG. 8B is another illustration of the medical system of FIG. 8A.

[0021] FIG. 8C is a further illustration of the medical system of FIG. 8A and FIG. 8B.

[0022] FIG. 9 is a schematic illustration of an implantable medical device.

[0023] FIG. 10 illustrates an example technique for positioning a medical system within a heart.

DETAILED DESCRIPTION

[0024] This disclosure describes a medical system configured to position one or more electrodes within a chamber of a heart. The medical system includes a device head configured to be positioned within the atrium and an implantable medical device (“IMD”) configured to be positioned within a vena cava leading to the atrium. A flexible lead extends between the device

head and the IMD. The IMD includes therapy delivery circuitry configured to provide stimulation signals via the flexible lead to the one or more electrodes of the device head. The flexible lead is configured to allow the device head to engage tissue within the atrium (e.g., in the triangle of Koch region or another location) as the IMD is positioned within the vena cava. In examples, the lead is configured to allow the device head to engage tissues within a right atrium of a heart as the IMD is positioned within inferior vena cava (IVC) of the heart. Positioning the IMD within the vena cava may improve spacing constraints compared to systems configured for positioning an IMD within the atrium itself. For example, positioning within the vena cava may allow the IMD to be larger, which may allow increased battery capacity, circuit complexity, and/or other improvements compared to IMDs configured for positioning within an atrium of a heart.

[0025] The lead extending between the device head and the IMD is configured to exhibit sufficient flexibility to allow a clinician to position the device head at a desired target site (e.g., The triangle of Koch region, or another target site on the atrial tissue) as the IMD remains positioned substantially within the vena cava (e.g., within the IVC). For example, a body of the device head may define an axis (“head axis”) while the body of the IMD defines a separate axis (“IMD axis”). The lead may be configured to exhibit a flexibility allowing the head axis to substantially maintain a plurality of orientations relative to the IMD axis, such that the device head may be maneuvered toward a target site within the atrium as the IMD remains substantially within the vena cava. In examples, the lead exhibits sufficient flexibility to extend between an IMD in an IVC to a device head positioned to place an electrode in proximity to (e.g., in contact with) a target site within a triangle of Koch region. The lead may be sufficiently flexible to allow a plurality of relative orientations between the head axis and the IMD axis as the device head is caused (e.g., by a clinician) to negotiate a path from a vena cava to a target site in the atrium (e.g., a target site in the triangle of Koch region).

[0026] The medical system may be configured to substantially maintain the one or more electrodes of device head in close proximity to a target site within the atrium as the IMD remains substantially within the vena cava. In examples, a fixation element configured to engage tissue is attached to the device head. The fixation element may be configured to engage tissue (e.g., atrial tissue) to secure the device head and IMD to tissue and to place the one or more electrodes of the device head in a desired engagement with tissue at the target site (e.g., in an atrium). The fixation element may include, for example, one or more fixation tines configured to extend distally from the device head. The one or more electrodes are configured to receive stimulation signals from the IMD and deliver sensing/therapy signals to the tissue based on the stimulation signals. In examples, the one or more electrodes include a shallow electrode configured to deliver sensing/therapy signals to tissue (e.g., atrial tissue) in a substantially non-invasive manner. In examples, the one or more electrodes may include a deep electrode configured to be advanced into the tissue for sensing signal from and the delivery of therapy signals to tissue such as ventricular tissue, or tissue of the His-Purkinje conduction system. The device head may include one or more shallow electrodes, one or more deep electrodes, and/or a combination of shallow electrodes and deep electrodes. The medical system may include one or more return electrodes configured to be used with the shallow electrode and/or deep electrode for sensing and/or delivery of therapy. In some examples, the medical system may include an IMD fixation device configured to engage tissue to substantially maintain the IMD within the vena cava.

[0027] In examples, the medical system includes a delivery catheter defining a lumen configured to allow passage of the IMD, the lead, and the device head therethrough. The catheter may be configured to deliver the medical system intravenously through the vena cava. In examples, the catheter includes a proximal section and a distal section. The distal section may be configured to extend into the atrium (e.g., a right atrium) as the proximal section remains substantially within the vena cava (e.g., an IVC). The distal section may be configured to define a curvature relative to the proximal section when the distal section is in a substantially relaxed configuration (e.g., in the

substantial absence of external forces acting on the distal section). In examples, the distal section includes a shape memory alloy configured to cause the distal section of define the curvature. In some examples, the distal section is configured to define a first curvature and define a second curvature out of plane with the first curvature in order to, for example, cause the lumen to define a path from a vena cave to a target site within an atrium. For example, the distal section may be configured to cause the lumen to define a path from an IVC to a target site within the triangle of Koch region of a right atrium. In examples, the device head defines a shape (e.g., an outer perimeter) configured to mechanically engage with an inner surface of the catheter lumen as the device head passes through the catheter lumen, The device head may be configured to mechanically engage the inner surface to substantially maintain the device head in an orientation whereby the fixation mechanism of the device head substantially faces a lumen opening defined by the distal portion of the catheter.

[0028] The delivery catheter defines a distal opening into the lumen at a distal end of the delivery catheter. The device head, the lead, and the IMD are configured to translate (e.g., slidably translate) within the lumen and through the distal opening. Hence, the delivery catheter may be configured to define a path from a vena cava to a target site to position the device head, the lead, and the IMD. In examples, the lead extending between the device head and the IMD has a sufficient axial stiffness such that a force (e.g., a pushing and/or pulling force) imparted to the IMD (e.g., by a stylet) causes the lead to transmit at least some portion of the imparted force to the device head, causing movement of the IMD, the lead, and the device head through the lumen of the delivery catheter. In examples, the lead has a sufficient axial stiffness such that that a force (e.g., a pushing and/or pulling force) imparted to the device head (e.g., by a stylet) causes the lead to transmit at least some portion of the imparted force to the IMD, causing movement of the IMD, the lead, and the device head through the lumen of the delivery catheter.

[0029] FIG. 1 is a conceptual diagram illustrating an example medical system 10. Medical system 10 includes a device head 12, an implantable medical device 14 (“IMD 14”), and a lead 16 extending between device head 12 and IMD 14. In FIG. 1, device head 12, IMD 14, and lead 16 are positioned within a lumen 20 defined by delivery catheter 18 and shown with hidden lines. Delivery catheter 18 is configured to define a pathway through lumen 20 substantially from a vena cava of heart 26 (e.g., the IVC) to a chamber of heart 26 (e.g., the RA). Device head 12, lead 16, and IMD 14 are configured to be translated through lumen 20 to position device head 12 in proximity to a target site 32 within the chamber of heart 26 (e.g., the RA) as IMD 14 remains substantially within a vena cava of heart 26 (e.g., the IVC). Although the examples herein discuss delivery and placement of IMD 14 within the RA of heart 26, medical system 10 may be configured to position IMD 14 in other chambers of heart 26 in a like manner as that described for the RA of heart 26.

[0030] Device head 12, lead 16, and IMD 14 may be configured to translate collectively through lumen 20 in the distal direction D and/or the proximal direction P. For example, device head 12, lead 16, and IMD 14 may be configured such that a force on device head 12 or IMD 14 causes motion of both device head 12 and IMD 14 through lumen 20. In examples, IMD 14 includes a distal end 15 (“IMD distal end 15”) and a proximal end 17 (“IMD proximal end 17”). Device head 12 may include a first side 11 (“head first side 11”) and a second side 13 (“head second side 13”) opposite head first side 11. Lead 16 may be coupled to IMD distal end 15 and head first side 11. In other examples, lead 16 may be coupled to other portions of IMD 14 and/or device head 12 to extend between IMD 14 and device head 12.

[0031] An electrode 19 is coupled to device head 12. Electrode 19 is electrically connected to circuitry within IMD 14 via lead 16, and thereby configured to receive stimulation signals from IMD 14 for delivery to tissues (e.g., atrial tissue) of heart 26, as well as allow IMD 14 to sense electrical signals from heart 26. In examples, electrode 19 is coupled to head second side 13. In examples, as depicted in FIG. 1, electrode 19 is a shallow electrode configured for sensing and

delivery of therapy signals to tissue in a substantially non-invasive manner. In other examples, electrode **19** may be a deep electrode configured to penetrate the tissues for sensing and delivery of therapy signals. Medical system **10** may include any number of electrodes in any configuration, including one or more electrodes supported by device head **12**, IMD **14**, and/or lead **16**.

[0032] A fixation element **21** configured to engage tissue is attached to device head **12**. Fixation element **21** may include, for example, one or more elongated tines such as fixation tine **23** configured to substantially maintain an orientation of device head **12** with respect to target site **32**. Fixation element **21** may include fixation tines of any shape, including helically-shaped fixation tines. Fixation element **21** (e.g., fixation tine **23**) may be configured to penetrate tissue in the vicinity of target site **32** to substantially maintain the orientation. In examples, fixation element **21** is configured to substantially maintain an orientation of electrode **19** with respect to target site **32** when fixation element **21** engages tissue. In examples, fixation element **21** may be configured to substantially maintain contact between electrode **19** and tissues within target site **32** when fixation element **21** engages the tissue.

[0033] Lead **16** extends between device head **12** and IMD **14**. As will be discussed, lead **16** is configured to extend between device head **12** and IMD **14** when device head **12** is positioned substantially within an atrium of heart **26** (e.g., the RA) and IMD **14** is positioned substantially within a vena cava of heart **26** (e.g., the IVC). Lead **16** may be a flexible lead configured to accommodate curvatures necessary when IMD **14** is positioned substantially within a vena cava of heart **26** and device head **12** is positioned substantially within an atrium of heart **26**. In examples, a distal end **25** of lead **16** (“lead distal end **25**”) may be coupled to device head **12**. A proximal end **27** of lead **16** (“lead proximal end **27**”) may be coupled to IMD **14**. In examples, lead **16** is coupled to IMD distal end **15** and head first side **11**. Lead **16** may be configured to enable transmission of stimulation signals from circuitry of IMD **14** to electrode **19** and/or sensing signals from electrode **19** to circuitry of IMD **14**. Lead **16** may include an electrical conductor configured to transmit the stimulation signals and/or sensing signals.

[0034] Lead **16** may be configured to cause movement of medical system **10** when a force (e.g., in the distal direction D and/or proximal direction P) is exerted on IMD **14** and/or device head **12**. Lead **16** may be configured to transmit a force exerted on one of IMD **14** or device head **12** (e.g., in the distal direction D or proximal direction P) to the other of IMD **14** or device head **12**. For example, lead **16** may be configured such that when IMD **14** exerts a force in the distal direction D or proximal direction P on lead proximal end **27**, lead distal end **25** exerts a force in the same direction on device head **12**. Lead **16** may be configured such that when device head **12** exerts a force in the distal direction D or proximal direction P on lead distal end **25**, lead proximal end **27** exerts a force in the same direction on IMD **14**.

[0035] In examples, lead **16** defines a lead axis from lead proximal end **27** to lead distal end **25**. The lead axis may linear, curved, and/or curvilinear. Lead **16** may be configured to exhibit a bending moment about the lead axis, such that lead **16** may transmit an axial force (e.g., between IMD **14** and device head **12**) as lead **16** accommodates any curvatures necessary to extend between device head **12** and IMD **14**. Lead **16** may be configured to transmit a force exerted on one of IMD **14** or device head **12** to the other of IMD **14** or device head **12** when lead axis is substantially linear, curved, and/or curvilinear.

[0036] Delivery catheter **18** defines lumen **20** extending through a proximal section **22** (“catheter proximal section **22**”) and a distal section **24** (“catheter distal section **24**”). Lumen **20** extends through catheter proximal section **22** and catheter distal section **24** to a lumen opening **28** defined at a distal end **30** of delivery catheter **18** (“catheter distal end **30**”). An inner wall **29** of delivery catheter **18** may define lumen **20** and lumen opening **28**. Delivery catheter **18** may define an axis Lc extending through lumen **20** and lumen opening **28**. Delivery catheter axis Lc may be substantially linear, curved, and/or curvilinear.

[0037] Delivery catheter **18** may be configured to allow a clinician to deliver medical system **10**

through a vena cava (e.g., the IVC) for implantation of device head **12** within an atrium (e.g., the RA) of heart **26**. Delivery catheter **18** may be intravenously transited through the IVC such that catheter distal section **24** passes into the RA. Delivery catheter **18** may be configured to substantially deliver medical system **10** (e.g., device head **12**) to the target site **32**. In examples, delivery catheter **18** is configured to allow the clinician to position catheter distal end **30** in proximity to a target site **32** as catheter distal section **24** transitions from the vena cava (e.g., the IVC) into the atrium (e.g., RA). Target site **32** may include an appendage or triangle of Koch region of the RA, or some other portion of heart **26**, or other locations within a body of a patient. In examples, delivery catheter **18** is configured to substantially position catheter distal end **30** such that lumen opening **28** substantially faces target site **32** (e.g., such that a vector normal to lumen opening **28** intersects target site **32**). In examples, catheter distal section **24** is configured to position lumen opening **28** in proximity to target site **32** in an atrium of heart **26** (e.g., the RA) when catheter distal section **24** substantially resides in a vena cava of heart **26** (e.g., the IVC).

[0038] Device head **12**, IMD **14**, and lead **16** may be positioned within any portion of lumen **20**. In examples, device head **12** and at least a portion of lead **16** are configured to reside within catheter distal section **24** as catheter distal end **30** is positioned in the proximity of target site **32**. In examples, IMD **14** is configured to reside within catheter proximal section **22** as catheter distal end **30** is positioned in the proximity of target site **32**. In some examples, lead **16** is configured to extend from catheter distal section **24** to catheter proximal section **22** as catheter distal end **30** is positioned in the proximity of target site **32**. In some examples, device head **12**, IMD **14**, and lead **16** are configured to reside outside of delivery catheter **18** as catheter distal end **30** is positioned in the proximity of target site **32**, and configured to be inserted into lumen **20** at a proximal end of delivery catheter **18** when catheter distal end **30** is positioned in the proximity of target site **32**. Device head **12**, IMD **14**, and lead **16** are configured to be translatable through at least some portion of lumen **20** when catheter distal end **30** is positioned in the proximity of target site **32**. In examples, device head **12** defines a shape (e.g., an outer perimeter **C**) configured to mechanically engage with an inner surface defining lumen **20** as device head **12** passes through the lumen **20**. Device head **12** may be configured to mechanically engage the inner surface to substantially maintain device head **12** in an orientation whereby second side **13** substantially faces lumen opening **28** as device head **12** translates through delivery catheter **18**.

[0039] FIG. 2 is a conceptual diagram illustrating delivery catheter **18** having positioned catheter distal end **30** in the proximity of target site **32**. Medical system **10** is positioned within delivery catheter **18** such that device head **12** is positioned substantially at or near lumen opening **28**. Device head **12** and some portion of lead **16** are within catheter distal section **24** while IMD **14** is substantially within catheter distal section **24**. Medical system **10** may be caused to translate as delivery catheter **18** translates using, for example, a force on medical system **10** exerted by an elongated member **38** (e.g., a stylet) extending within lumen **20** of delivery catheter **18**. In FIG. 2, catheter proximal section **22** is substantially within a vena cava (e.g., the IVC) of a heart **26**. Catheter distal section **24** is substantially within an atrium (e.g. the RA) of heart **26**.

[0040] Delivery catheter **18** may be configured to position in the atrium such that lumen opening **28** substantially faces target site **32**. Target site **32** may be, for example, an area within the triangle of Koch region of heart **26**, or some other target area within heart **26**. In examples, delivery catheter **18** defines a first curvature **34** and/or a second curvature **36**, in order to assist the clinician in positioning catheter distal end **30** in the proximity of target site **32**. Delivery catheter **18** (e.g., catheter distal section **24**) may define the first curvature **34** and/or second curvature **36** when catheter distal section **24** is in a substantially relaxed configuration (e.g., in the substantial absence of external forces acting on distal section **24**). In examples, first curvature **34** and/or second curvature **36** are configured to substantially define a pathway through lumen **20** from the IVC (e.g., from the inferior end of the IVC) to target site **32** in the RA of heart **26**. Delivery catheter **18** may be configured such that first curvature **34** defines a curvature in a first plane (e.g., a first Euclidean

plane) and second curvature **36** defines a curvature in a second plane (e.g., a second Euclidean plane). The first plane and the second plane may define an angle between the first plane and the second plane when delivery catheter **18** defines first curvature **34** and second curvature **36**, such that first curvature **34** and second curvature **36** are out-of-plane with one another. In examples, delivery catheter **18** may be configured to allow a clinician to establish contact between a portion of catheter distal section **24** defining lumen opening **28** and tissues (e.g., atrial tissue) within target site **32**.

[0041] In examples, IMD **14** is configured to receive a force in the distal direction D (e.g., a pushing force) to cause medical system **10** to translate through lumen **20** in the distal direction D. IMD **14** may be configured to receive a force in the proximal direction P (e.g., a pulling force) and cause medical system **10** to translate through lumen **20** in the proximal direction P. IMD **14** may be configured to engage elongated member **38** (e.g., a stylet) to receive the force in the distal direction D and/or proximal direction P to, for example, cause medical system **10** (e.g., IMD **14**, lead **16**, and device head **12**) to move in the same direction as the received force. In some examples, device head **12** is configured to receive a force (e.g., from a stylet) in the distal direction D (e.g., a pushing force) and/or proximal direction P (e.g., a pulling force) to cause medical system **10** to translate through lumen **20** in the direction of the exerted force.

[0042] Thus, delivery catheter **18** may be configured to substantially guide device head **12** toward or away from catheter distal end **30** when a force is exerted on medical system **10** (e.g., on IMD **14** and/or device head **12**). Delivery catheter **18** may be configured to substantially provide a pathway through lumen **20** to catheter distal end **30** for the translation of medical system **10**, such that a clinician may cause medical system **10** (e.g., device head **12**) to position at or near lumen opening **28** by exerting the force in the distal direction on medical system **10**.

[0043] Delivery catheter **18** defines lumen opening **28** such that device head **12**, fixation element **21**, electrode **19**, lead **16**, and IMD **14** may pass therethrough. Fixation element **21** may be configured to engage tissue within target site **32** as fixation element **21** passes through lumen opening **28**. In examples, fixation element **21** (e.g., fixation tine **23**) is configured to extend distally from device head **12** as medical system **10** translates distally through catheter distal section **24**. Fixation element **21** may be configured to penetrate tissues within target site **32** as fixation element **21** passes through lumen opening **28** in order to engage the tissues. For example, a portion of fixation element **21** (e.g., fixation tine **23**) may be resiliently biased to expand outward as fixation element **21** passes through lumen opening **28**, in order to aid in grasping the tissue. Delivery catheter **18** may be configured to radially constrain fixation element **21** (e.g., fixation tine **23**) as device head **12** translates within delivery catheter **18** toward or away from lumen opening **28**.

[0044] Catheter distal section **24** is configured to maintain device head **12** in an orientation such that some portion of fixation element **21** (e.g., a free end of fixation tine **23**) passes through lumen opening **28** before device head **12** passes through lumen opening **28**. For example, when fixation elements **21** (e.g., fixation tine **23**) extends distally from head second side **13**, catheter distal section **24** may be configured such that head second side **13** substantially faces lumen opening **28** and head first side **11** substantially faces away from lumen opening **28** as device head **12** translates within catheter distal section **24** toward lumen opening **28**. Hence, catheter distal section **24** and medical system **10** may be configured such that fixation element **21** engages tissues at target site **32** when catheter distal section **24** guides medical system **10** (e.g., device head **12**) toward lumen opening **28** substantially facing target site **32**. Additionally, with lumen opening **28** configured to allow with device head **12**, lead **16**, and IMD **14** to pass therethrough, delivery catheter **18** may be withdrawn proximally from the atrium and the vena cava as device head **12** substantially remain in the atrium (e.g., the RA), and IMD **14** remain substantially with the vena cava (e.g., the IVC). Delivery catheter **18** may be configured to substantially straighten when proximally withdrawn to, for example, allow device head **12**, lead **16**, and IMD **14** to pass through lumen opening **28**. For example, catheter distal section **24** may be configured to reduce and or substantially eliminate a

curvature of first curvature **34** and/or second curvature **36** to, for example, allow passage of catheter distal section **24** over IMD **114**.

[0045] FIG. **3** is a conceptual diagram illustrating device head **12** within the RA of heart **26** as IMD **14** resides substantially within the IVC of heart **26**. Delivery catheter **18** has been proximally withdrawn from the RA and IVC. Fixation element **21** (e.g., fixation tine **23**) is grasping tissues within target site **32** to substantially maintain device head **12** within the RA. Lead **16** extends between device head **12** and IMD **14** substantially within the IVC. Medical system **10** is configured such that IMD **14** may transmit stimulation signals to an electrode (e.g., electrode **19** (FIG. **1**)) coupled to device head **12**. Fixation element **21** causes device head **12** to substantially maintain a position whereby the electrode may deliver therapy signals to tissues within target site **32** based on the stimulation signals received from IMD **14** via lead **16**, and IMD **14** may sense signals from the tissues via the electrode. In examples, medical system **10** includes an IMD fixation device **33** configured to engage tissue to substantially maintain IMD **14** within the vena cava when device **12** positions in the RA (or another chamber of the heart). Fixation device **33** may include, for example, one or more resiliently biased members configured to expand outward when delivery catheter **18** is withdrawn proximally, a mechanism (e.g., a wire or other mechanism) configured to be operated by a clinician to cause fixation device **33** to engage the tissue, or some other structure sufficient to engage tissue to substantially maintain IMD **114** within the vena cava.

[0046] FIG. **4** illustrates an example medical system **100**. Medical system **100** is an example of medical system **10**. Medical system **100** includes device head **112** including head first side **111** and head second side **113**, IMD **114** including IMD distal end **115** and IMD proximal end **117**, lead **116** including lead distal end **125** and lead proximal end **127**, fixation element **121** including fixation tine **123**, and IMD fixation device **133**, all of which may be configured individually and in relation to each other in the same manner as that discussed for the like-named components of medical system **10**. In FIG. **4**, device head **112** defines an axis Lh intersecting head first side **111** and head second side **113**. IMD **114** define an axis Li intersecting IMD distal end **115** and IMD proximal end **117**.

[0047] Medical system **100** may include one or more electrodes configured to deliver low-voltage electrical pulses to the heart or sense a cardiac electrical activity, e.g., depolarization and repolarization of the heart. The one or more electrodes may be coupled to device **112**. For example, in FIG. **4**, shallow electrode **140** and deep electrode **142** are coupled to second side **113** of device head **112**. Shallow electrode **140** and/or deep electrode **142** may be examples of electrode **19** (FIG. **1**). Shallow electrode **140** may be configured to deliver sensing/therapy signals to tissue in a substantially non-invasive manner. In examples, shallow electrode **140** is an atrial electrode. Deep electrode **142** may be configured to be advanced into tissue for the delivery of sensing/therapy signals. In some examples, deep electrode **142** is supported by an electrode support member **141** extending distally from device head **112**. Electrode support member **141** and/or deep electrode **142** may be configured to penetrate tissues to substantially implant deep electrode **142** within the tissues (e.g., to target the His bundle (HB), right bundle branch (RBB), left bundle branch (LBB), or other ventricular tissue) Shallow electrode **140** and/or deep electrode **142** may be configured to receive stimulations signals from processing circuitry within IMD **114**. Shallow electrode **140** and/or deep electrode **142** may be any of a number of different types of electrodes, including button electrodes, pad electrodes, helical electrodes, tine electrodes, or the like.

[0048] Fixation element **121** may be configured to engage tissue and substantially maintain an orientation of device head **112** relative to the tissue. Fixation element **121** includes one or more elongated tines such as fixation tine **123**. Fixation tine **123** is an example of fixation tine **23** (FIGS. **1-3**). In general, a tine may refer to any structure that is capable of securing a lead or leadless implantable medical device to a location within the heart. In some examples, a tine (e.g. fixation tine **123** and/or electrode support member **141**) may be composed of a shape-memory alloy that allows deformation along the length of the tine. A tine may be substantially flat along the length of

the tine. In other examples, a tine may substantially define a helix and/or helical member.

[0049] In examples, fixation tine **123** includes a fixed end **146** coupled to device head **112** and a free end **148** opposite fixed end **146**. In examples, free end **148** is configured to penetrate tissue. Fixation tine **123** may be biased so that at least some portion of fixation tine **123** expands radially as fixation tine **123** passes through lumen opening **28** (FIG. 1). Fixation tine **123** may be biased to drive free end **148** radially outward from device head axis Lh as free end **148** passes through lumen opening **28**. The biasing tending to drive free end **148** radially outward as fixation tine **123** extends through lumen opening **28** may cause fixation tine **123** to substantially grasp tissue and more securely anchor device head **112** within an atrium of heart **26**. Free end **148** may pierce the tissue and may act to pull device head **112** and other portions of medical system **10** toward a target site (e.g., target site **32** (FIG. 1)) as fixation tine **123** elastically bends or curves radially outward. Fixation element **121** may include any number of fixation tines, which may be configured similarly to fixation tine **123**.

[0050] The biasing of fixation tine **123** tending to drive free end **148** radially outward may cause fixation tine **123** to assume any general shape. In some examples, the biasing of fixation tine **123** tends to cause fixation tine **123** to position such that free end **148** establishes a position proximal to a midpoint M between fixed end **146** and free end **148** (e.g., as depicted at FIG. 4). In some examples, the biasing of fixation tine **123** tends to cause fixation tine **123** to position such that free end **148** remains distal to midpoint M between fixed end **146** and free end **148**. Fixation tine **123** may be formed to have a preset shape and may be formed using any suitable material. In examples, fixation tine **123** comprises a nickel-titanium alloy such as Nitinol.

[0051] In some examples, fixation tine **123** may be configured to substantially maintain a delivery configuration where free end **148** is distal to fixed end **146** and distal to midpoint M (e.g., as depicted in FIG. 1). For example, fixation tine **123** may be configured to substantially maintain the delivery configuration when free end **148** is constrained from outward radial motion by inner wall **29** of delivery catheter **18** (FIGS. 1-2). Delivery catheter **18** may be configured to substantially maintain fixation tine **123** in the delivery configuration as device head **112** translates within lumen **20** of delivery catheter **18**. Substantially maintaining free end **148** distal to midpoint M (e.g., in the delivery configuration) may facilitate the penetration of tissue by free end **148** when fixation tine **123** passes through lumen opening **28** of delivery catheter **18**.

[0052] Device head **112** may be configured to limit its rotation when within a lumen defined by a delivery catheter (e.g., lumen **20** defined by delivery catheter **18** (FIGS. 1-2)). For example, device head **112** may be configured to limit its rotation such that head second side **113** remains distal to head first side **111** as device head **112** translates distally and/or proximally within lumen **20**. In some examples, device head **112** is configured to maintain some portion of fixation element **121** distal to head second side **113** as device head **112** translates within lumen **20**, in order to cause the portion of fixation element **121** to pass through lumen opening **28** before device head **112** passes through lumen opening **28**. Substantially maintaining the portion of fixation element **121** distal to device head **112** as device head **112** translates distally toward lumen opening **28** (FIGS. 1-3) may assist in the engagement of fixation element **121** with tissues within target site **32** (FIGS. 1-3), such that fixation element **121** may substantially maintain an orientation of device head **112** with respect to target site **32**.

[0053] In examples, device head **112** may define a dimension substantially perpendicular to device head axis Lh, with the dimension configured to cause some portion of device head **112** to engage an inner wall (e.g., inner wall **29**) of the lumen, in order to limit the rotation of device head axis Lh relative to the inner wall. For example, device head **112** may define a dimension configured to limit a rotation of device head axis Lh relative to a portion of catheter axis Lc (FIGS. 1-2). Delivery head **112** may include one or more protrusions **150** extending radially outward from device head axis Lh and defining the dimension. Protrusion **150** may be configured to engage inner wall **29** of delivery catheter **18** (FIGS. 1-2) when device head **112** causes (or attempts to cause) rotation of device head

axis Lh relative to catheter axis Lc. Device head **112** may include any number of protrusions **150**, and protrusion **150** may define any geometry. In other examples, in addition to or instead of protrusion **150**, device head **112** may define a substantially constant radius around device head axis Lh configured to limit a rotation of device head axis Lh relative to a delivery catheter axis Lc. [0054] Lead **116** extends between device head **112** and IMD **114**. In examples, lead distal end **125** is coupled to head first side **111** and lead proximal end **127** is coupled to IMD distal end **115**. Lead **116** is configured to cause movement of distal head **112** and IMD **114** when a force in the distal direction D or proximal direction P is imparted to distal head **112** or IMD **114** (e.g., by stylet **38** (FIG. 2)). Lead **116** may be configured to transmit a distal and/or proximal force from IMD **114** to device head **112**, and/or from device head **112** to IMD **114**.

[0055] In examples, lead **116** is configured such that movement of head first side **111** causes movement of lead distal end **125**, and vice-versa. In examples, lead **116** is configured such that movement of IMD distal end **115** causes movement of lead proximal end **127**, and vice-versa. Lead **116** may be configured such that a movement of IMD **114** or device head **112** in the distal direction D causes the other of IMD **114** or device head **112** to move in the distal direction D. Lead **116** may be configured such that a movement of IMD **114** or device head **112** in the proximal direction P causes the other of IMD **114** or device head **112** to move in the proximal direction P. In examples, lead **116** is configured to transmit the force between IMD **114** and device head **112** at least partially through a lead axis running through lead **116** from lead proximal end **127** to lead distal end **125**. Lead **116** may be configured to transmit the force when the lead axis defines a linear, curved, and/or curvilinear shape.

[0056] In examples, lead **116** is configured to substantially push device head **112** when lead **116** receives a distal force (e.g., a force in the distal direction D) from IMD **114**. Lead **116** may be configured such that, when lead proximal end **127** receives the distal force from IMD **114**, lead distal end **125** transmits some portion of the force to device head **112**. Lead **116** may be configured to transmit the portion of the distal force to device head **112** when the lead axis defines one or more curvatures, such as when lead **116** extends through one or more of first curvature **34** and/or second curvature **36** (FIGS. 1-2). In examples, lead **116** includes an outer lead body **152** configured to transmit the distal force to device head **112** when lead **116** defines the one or more curvatures. For example, outer lead body **152** may include a helical coil substantially surrounding some or all of the lead axis between lead proximal end **127** and lead distal end **125**. In some examples, outer lead body **152** may be a substantially unitary body having one or more slits at least partially around a circumference of outer lead body and configured to allow bending of lead **116** while transferring a distal force from lead proximal end **127** to lead distal end **125**. Outer lead body may have other configurations in other examples.

[0057] Lead **116** may be a flexible lead configured to allow positioning of device head **112** relative to IMD **114** in a variety of relative orientations. Lead **116** may be configured (e.g., have sufficient flexibility) to allow device head axis Lh to rotate and/or translate relative to IMD axis Li as device head **112** and IMD **114** translates through lumen **20** of delivery catheter **18** (FIGS. 1-2)). In examples, lead **116** is configured to allow device head axis Lh and IMD axis Li to remain substantially parallel with catheter axis Lc (FIGS. 1-2) as device head **112** and IMD **114** translate through lumen **20** of delivery catheter **18**. For example, lead **116** may be configured to allow device head axis Lh to remain substantially parallel with a portion of catheter axis Lc in catheter distal section **24** as device head **112** translates through catheter distal section **24** (e.g., through portions defining first curvature **34** and/or second curvature **36** (FIGS. 1-2)). Lead **116** may be configured to allow IMD axis Li to remain substantially parallel with a portion of catheter axis Lc in catheter proximal portion **24** as device head **112** translates through catheter distal section **24**.

[0058] In examples, lead **116** is configured to extend between device head **112** and IMD **114** when fixation element **121** engages tissue within an atrium (e.g., the RA) of heart **26** and IMD **114** is positioned substantially within a vena cava (e.g., the IVC) of heart **26**. Lead **116** may be configured

to have sufficient flexibility to extend between device head **112** and IMD **114** when device head axis Lh and IMD axis Li define skew lines (e.g., non-intersecting and non-parallel lines), substantially parallel lines, and/or intersecting lines. Lead **116** may be configured to have sufficient flexibility such that the lead axis of lead **116** may define one or more curvatures relative to IMD axis Li and/or device head axis Lh. In examples, lead **116** may be configured to define a primary curvature and a secondary curvature, with the primary curvature defining a curvature in a primary plane (e.g., a primary Euclidean plane) and the secondary curvature defining a curvature in a secondary plane (e.g., a secondary Euclidean plane). The primary plane and the secondary plane may define an angle between the primary plane and the secondary plane when lead **116** defines primary curvature and secondary curvature, such that the primary curvature and secondary curvature are out-of-plane with one another.

[0059] Lead **116** may be configured to enable transmission of stimulation signals from IMD **114** to electrode **140** and/or electrode **142** and/or sensing signals from electrode **140** and/or electrode **142** to IMD **14**. Lead **116** may include an electrical conductor (not shown) configured to transmit the stimulation signals and/or sensing signals. In examples, the electrical conductor is surrounded by outer lead body **152** of lead **116**. In some examples, outer lead body **152** comprises the electrical conductor. The electrical conductor may be configured to establish electrical connectivity between one or more electrodes coupled to device head **112** (e.g., electrode **140** and/or electrode **142**) and processing circuitry within IMD **114**. The conductor may include an insulative jacket around all or some portion of the conductor. The insulative jacket may electrically isolate the conductor from other portions of medical system **100** as well as from the environment within the patient.

[0060] In examples, the conductor is a multi-conductor comprising a plurality of conductors. The conductor may be configured such that each individual conductor in the plurality of conductors is electrically connected to an individual electrode (e.g., shallow electrode **140** or deep electrode **142**) in a plurality of electrodes coupled to device head **112**, IMD **114**, and/or lead **116**. The conductor may be configured such that each individual conductor is electrically connected to processing and/or sensing circuitry within IMD **114**. The individual conductor may be configured to conduct respective signals between circuitry of IMD **114** and individual electrodes. Hence, medical system **100** may be configured to, for example, provide multi-point stimulation to tissue at or in the vicinity of a target site in a patient (e.g., target site **32** (FIGS. 1-3)).

[0061] IMD **114** may be configured to reside substantially within a vena cava of heart **26** (e.g., the IVC) when device head **112** and portions of lead **116** reside substantially within an atrium of heart **26** (e.g., the RA). IMD **114** may include a housing **154** (“IMD housing **154**”). One or more portions of IMD housing **154** may define IMD distal end **115** and IMD proximal end **117**. Medical system **100** may include one or more return electrodes such as return electrode **156** and return electrode **157** configured to deliver sensing/therapy in tandem with other electrodes of medical system **10**. Return electrodes **156**, **157** may be at any location on medical system **10**. In examples, return electrode **156** is coupled to IMD housing **154** and configured to deliver the sensing/therapy in tandem with one or more electrodes coupled to device head **112** (e.g., shallow electrode **140** and/or deep electrode **142**). In examples, IMD **154**, device head **112**, and/or lead **116** may include a nonconductive coating. Return electrodes **156**, **157** and/or other electrodes of medical system **100** may be defined as an uncoated portion of medical system **100**. For example, IMD housing **154** may include a nonconductive coating and define return electrode **156** as an uncoated portion of IMD housing **154**.

[0062] IMD housing **154** may enclose processing and/or sensing circuitry (not shown) configured to deliver stimulation signals to and/or receive sensing signals from shallow electrode **140**, deep electrode **142**, and/or return electrode **156**. The processing and/or sensing circuitry may be configured to provide electrical signals, e.g., pacing therapy, to shallow electrode **140**, deep electrode **142**, and/or return electrode **156**. The processing and/or sensing circuitry may be configured to receive electrical signals, e.g., sensed cardiac electrical signals, from shallow

electrode **140**, deep electrode **142**, and/or return electrode **156**. IMD housing **154** may be configured to fluidly isolate the processing and/or sensing circuitry from an environment in contact with an exterior surface of IMD housing **154**. In examples, IMD housing **154** is configured to hermetically seal an enclosure defined by IMD **114** and holding the processing and/or sensing circuitry. IMD housing **154** may be configured to define shapes that are easily accepted by the patient's body while minimizing patient discomfort. For example, IMD housing **154** may define a substantially cylindrical shape with cylindrical sidewalls. In other examples, IMD housing **154** may define substantially rectangular or other non-cylindrical shapes. IMD housing **154** may define shapes in which corners and edges are designed with relatively large radii, in order to present a housing having smoothly contoured exterior surfaces.

[0063] In examples, medical system **100** includes a fixation assembly (not shown) configured to assist in securing IMD **114** within a vena cava when delivery catheter **18** (FIGS. **1-2**) has been proximally withdrawn. The fixation assembly may be coupled to IMD **114** (e.g., IMD housing **154**) or some other portion of medical system **100**. The fixation assembly may be configured to assume a delivery configuration when medical system **100** resides within lumen **20** of delivery catheter **18** and a deployment configuration when medical system **100** is outside of lumen **20** (e.g., when delivery catheter **18** has been proximally withdrawn). In examples, the fixation assembly is configured to expand (e.g., radially expand) when transitioning from the delivery configuration to the deployment configuration. The fixation assembly may be configured to expand to cause engagement with a vessel wall, such as an interior vessel wall of a vena cava of heart **26**.

[0064] The fixation assembly may be configured to translate with IMD **114** when IMD **114** translates within lumen **20** of delivery catheter **18**. In some examples, the fixation assembly is configured to engage an actuation member such as a tether, with the actuation member configured to cause the fixation assembly to transition from the delivery configuration to the deployment configuration. The actuation mechanism (e.g., the tether) may be accessible from outside of the vasculature of a patient, in order to allow a clinician to operate the actuation mechanism and cause transition to the delivery configuration. For example, some portion of the actuation mechanism (e.g., a proximal end) may extend from a proximal opening of delivery catheter **18**.

[0065] The fixation assembly may be configured to engage a vessel wall of a vena cava in the deployment configuration. In examples, the fixation assembly is configured such that, on operation of the actuation member (e.g., the tether), the fixation assembly establishes physical contact with the vessel wall to maintain the positional integrity of one or more portions of medical system **100** (e.g., IMD **114**). The fixation assembly may be configured to engage the vessel wall in a manner minimizing obstruction to blood flow through the vena cava. The fixation assembly may be configured to substantially secure IMD **114** within a vena cava in a manner minimizing obstruction to blood flow through the vena cava. The fixation assembly may apply little more than the force that is appropriate to one or more portions of medical system **100** (e.g., IMD **114**) in place without applying excessive force to the vessel wall. The fixation assembly may be constructed to apply light, but sufficient, force to the vessel wall. Such forces are at least less than those associated with the placement of vascular stents in which the objective may be to press against the vascular wall with sufficient force to provide scaffolding support for the vessel wall.

[0066] In examples, the fixation assembly is coupled to IMD housing **154**. The fixation assembly may be configured to expand radially outward from IMD housing **154** to substantially secure IMD **114** within a vena cava of heart **26** when device head **112** resides within an atrium of heart **26**. In examples, the fixation assembly may include one or more loops coupled to IMD housing **154** and configured to expand to engage a vessel wall of a vena cava. The loops may be formed, for example, by an elongated member biased into a curved shape to form the one or more loops. The one or more loops may be configured to expand along an axis (e.g., a major axis and/or a minor axis of an ellipse defined by the loop) when transitioning from a delivery configuration to a deployed configuration. In some examples, one or more exterior surfaces of IMD housing **154** may

be sintered to promote tissue growth along the one or more exterior surfaces. Sintering the one or more exterior surfaces may reduce strain on the fixation assembly (e.g., as a result of the sintering providing some fixation force). In some examples, the fixation assembly may be configured to engage a vessel wall at one or more points around an inner periphery of the vessel wall. The fixation assembly may be configured to engage multiple points around the inner periphery in order to foster a resistance to rotation of IMD **114** around IMD axis **Li**, or for another reason.

[0067] FIG. 5A and FIG. 5B illustrate an example delivery catheter **218** configured to deliver device head **12**, **112** to a target site **32** within an atrium (e.g., the RA) of a heart **26**. Delivery catheter **218** is an example of delivery catheter **18** (FIGS. 1-2). Delivery catheter **218** includes lumen **220**, catheter proximal section **222**, catheter distal section **224**, lumen opening **228**, inner wall **229**, catheter distal end **230**, first curvature **234**, and second curvature **236**, which may be configured individually and in relation to each other in the same manner as that discussed for the like-named components of delivery catheter **18**.

[0068] Delivery catheter **218** may be configured to define first curvature **234** and/or second curvature **236** when medical system **10**, **100** (FIGS. 1-4) is positioned within delivery catheter **218**. For example, delivery catheter **218** may be configured to define first curvature **234** and/or second curvature **236** when device head **12**, **112** and/or portions of lead **16**, **116** are within catheter distal section **224**. Delivery catheter **218** defines a lumen **220** in a manner configured to allow passage of IMD **14**, **114**, lead **16**, **116**, and device head **12**, **112** therethrough. Lumen **220** extends through catheter proximal section **222** and catheter distal section **224**, and opens into lumen opening **228** at catheter distal end **230**. Catheter distal section **224** is configured to extend into an atrium of heart **26** (FIG. 1) as catheter proximal section **222** remains substantially within a vena cava of heart **26**. Lumen **220** and lumen opening **228** are sized to allow passage of device head **12**, **112**, lead **16**, **116**, and IMD **14**, **114** therethrough. Hence, delivery catheter **218** may be utilized to define a path using lumen **220** from a vena cava to a target site in an atrium for the positioning of medical system **10** within the atrium. Delivery catheter **218** is configured to be proximally withdrawn and to straighten to cause medical system **10** to pass through lumen opening **228**.

[0069] Delivery catheter **218** is illustrated in a substantially relaxed position, such that catheter distal section **224** is substantially free of external forces acting on catheter distal section **224**. Catheter distal section **224** is configured to define first curvature **234** and second curvature **236** in the substantially relaxed position. Catheter distal section **224** may be configured to define first curvature **234** out of plane with the second curvature **236** in order to, for example, cause lumen **220** to define a path from a vena cava to a target site within an atrium.

[0070] FIG. 6A and FIG. 6B illustrate an example delivery catheter **318** defining a pathway from a vena cava of heart **26** to an atrium of heart **26** using a lumen (e.g., lumen **20**, **220**) defined by delivery catheter **318**. Heart **26** is illustrated as transparent with delivery catheter **318** positioned inside heart **26** in FIG. 6A and FIG. 6B. Delivery catheter **318** includes catheter proximal section **322**, catheter distal section **324**, and catheter distal end **330**, which may be configured individually and in relation to each other in the same manner as that discussed for the like-named components of delivery catheter **18** and/or delivery catheter **218**. Delivery catheter **318** is an example of delivery catheter **18** and/or delivery catheter **218**.

[0071] In FIG. 6A and FIG. 6B, delivery catheter **318** is positioned such that catheter proximal section **322** is substantially within the vena cava of heart **26** and catheter distal section **324** is substantially within the atrium of heart **26**. FIG. 6A provides an anteroposterior (AP) view and FIG. 6B provides a view rotated 90 degrees from the AP view. Delivery catheter **318** (e.g., catheter distal section **324** and catheter proximal section **322**) may be inserted intravenously through the vena cava to cause catheter distal section **324** to enter the atrium. Delivery catheter **318** may be configured to define first curvature **34**, **234** and/or second curvature **36**, **236** as catheter distal section **324** enters the atrium, and may be configured to define first curvature **34**, **234** and/or second curvature **36**, **236** when catheter distal section **324** resides within the atrium.

[0072] FIG. 7 illustrates a delivery catheter **418** including catheter distal section **424**. Delivery catheter **418** is an example of delivery catheter **18**, delivery catheter **218**, and/or delivery catheter **318**. Delivery catheter **418** includes a lumen defining a pathway from a vena cava of heart **26** to an atrium of heart **26**. A medical system **400** including device head **412**, lead **416**, and IMD **414** is positioned within the lumen of delivery catheter **418**. Medical system **400** is an example of medical system **10** and/or medical system **100**. In FIG. 7, delivery catheter **418** is configured such that medical system **400** may be positioned within the lumen defined by delivery catheter **418** from a vena cava of heart **26** to an atrium of heart **26**. Delivery catheter **418** is configured to substantially guide device head **412** and portions of lead **416** from the vena cava to a target site **432** within the atrium. Device head **412**, lead **416**, and IMD **414** may be configured individually and relation to each other in the same manner as that discussed for the like-named components of medical system **10** and/or medical system **100**.

[0073] FIG. 8A, FIG. 8B, and FIG. 8C illustrate an example medical system **500** implanted within a heart **26**. Medical system **500** is an example of medical system **10**, medical system **100**, and/or medical system **400**. Heart **26** is illustrated as transparent with medical system **500** positioned inside heart **26** in FIG. 8A, FIG. 8B, and FIG. 8C. Medical system **500** includes device head **512**, lead **516**, IMD **514**, electrode **519**, and fixation element **521** including fixation tine **523**, all of which may be configured individually and relation to each other in the same manner as that discussed for the like-named components of medical system **10**, medical system **100**, and/or medical system **400**. Fixation element **521** is engaged with (e.g., grasping) tissues within a target site **532** and substantially maintaining device head **512** within an atrium of heart **26**. Lead **516** extends between device head **512** to IMD **514**. IMD **514** is positioned substantially within a vena cava of heart **26**. FIG. 8A provides an anteroposterior (AP) view and FIG. 8B provides a posteroanterior (PA) view. FIG. 8C provides a top view.

[0074] FIG. 9 is a functional block diagram illustrating an example configuration of IMD **14**, **114**, **414**, **514**. As shown in FIG. 9, IMD **14**, **114**, **414**, **514** includes processing circuitry **602**, sensing circuitry **604**, therapy delivery circuitry **606**, sensors **608**, communication circuitry **610**, and memory **612**. In some examples, memory **612** includes computer-readable instructions that, when executed by processing circuitry **602**, cause IMD **14**, **114**, **414**, **514** and processing circuitry **602** to perform various functions attributed to IMD **14**, **114**, **414**, **514** and processing circuitry **602** herein. Memory **612** may include any volatile, non-volatile, magnetic, optical, or electrical media, such as a random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other digital media.

[0075] Processing circuitry **602** may include fixed function circuitry and/or programmable processing circuitry. Processing circuitry **602** may include any one or more of a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or equivalent discrete or analog logic circuitry. In some examples, processing circuitry **602** may include multiple components, such as any combination of one or more microprocessors, one or more controllers, one or more DSPs, one or more ASICs, or one or more FPGAs, as well as other discrete or integrated logic circuitry. The functions attributed to processing circuitry **602** herein may be embodied as software, firmware, hardware or any combination thereof.

[0076] In some examples, processing circuitry **602** may receive (e.g., from an external device), via communication circuitry **610**, a respective value for each of a plurality of cardiac sensing parameters, cardiac therapy parameters (e.g., cardiac pacing parameters), and/or electrode vectors. Processing circuitry **602** may store such parameters and/or electrode vectors in memory **612**.

[0077] Therapy delivery circuitry **606** and sensing circuitry **604** are electrically coupled to electrodes **614**, which may correspond to electrode **19**, **519**, shallow electrode **140** and/or deep electrode **142**. Processing circuitry **602** is configured to control therapy delivery circuitry **606** to generate and deliver electrical therapy to heart **26** via electrodes **614**. Electrical therapy may

include, for example, pacing pulses, or any other suitable electrical stimulation. Processing circuitry **602** may control therapy delivery circuitry **606** to deliver electrical stimulation therapy via electrodes **614** according to one or more therapy parameter values, which may be stored in memory **612**. Therapy delivery circuitry **606** may include capacitors, current sources, and/or regulators, in some examples.

[0078] In addition, processing circuitry **602** is configured to control sensing circuitry **604** to monitor signals from electrodes **614** in order to monitor electrical activity of heart **26**. Sensing circuitry **604** may include circuits that acquire electrical signals, such as filters, amplifiers, and analog-to-digital circuitry. Electrical signals acquired by sensing circuitry **604** may include intrinsic and/or paced cardiac electrical activity, such as atrial depolarizations and/or ventricular depolarizations. Sensing circuitry **604** may filter, amplify, and digitize the acquired electrical signals to generate raw digital data. Processing circuitry **602** may receive the digitized data generated by sensing circuitry **604**. In some examples, processing circuitry **602** may perform various digital signal processing operations on the raw data, such as digital filtering. In some examples, in addition to sensing circuitry **604**, IMD **14**, **114**, **414**, **514** optionally may include sensors **608**, which may be one or more pressure sensors and/or one or more accelerometers, as examples. Communication circuitry **610** may include any suitable hardware (e.g., an antenna), firmware, software, or any combination thereof for communicating with another device, e.g., external to the patient.

[0079] A technique for positioning a medical system **100** within an atrium of a heart is illustrated in FIG. **10**. Although the technique is described mainly with reference to medical system **10** and delivery catheter **18** of FIGS. **1-3**, the technique may be applied to other medical systems in other examples, including medical systems **100**, **400**, **500** and delivery catheter **218**, **318**, **418**.

[0080] The technique includes positioning a device head **12** of medical system **10** within an atrium of a heart **26** by transmitting an axial force to device head **12** (**1002**). The technique may include positioning the device head **12** at a target site **32** within the atrium. The technique may include extending lead **16** from device head **12** to an IMD **14** within a vena cava of heart **26** (**1004**). The technique may include delivering sensing/therapy signals to tissues within target site **32** using an electrode **19** coupled to device head **12**. Electrode **19** may receive stimulation signals from IMD **14** and deliver the sensing/therapy signals to the tissue based on the stimulation signals. In examples, the technique includes sending the stimulation signals from processing circuitry within IMD **14** to electrode **19**.

[0081] In examples, the technique includes defining a pathway from the vena cava of heart **26** to target site **32** within the atrium of heart **26** using delivery catheter **18**. The technique may include translating delivery catheter **18** through the vena cava and causing some portion of delivery catheter **18** to enter the atrium. In examples, the technique includes causing catheter distal section **24** to enter the atrium. The technique may include defining a first curvature **34** and/or second curvature **36** using catheter distal section **24** when catheter distal section **24** positions or is positioned within the atrium. In examples, the technique includes positioning a distal end **30** of delivery catheter **18** in the vicinity of target site **32** using the first curvature **34** and/or the second curvature **36**. The technique may include defining the pathway from the vena cava to target site **32** using a lumen **20** defined by delivery catheter **18**.

[0082] The technique may include positioning medical system **10** within the atrium of heart **26** when delivery catheter **18** enters the atrium. The technique may include translating medical system **10** through lumen **20** to cause device head **12** to position within the atrium of heart **26**. In some examples, the technique includes causing medical system **10** to translate by exerting a distal or proximal force on IMD **14** (e.g., on IMD proximal end **17**). The technique may include transmitting some portion of the force on IMD **14** through lead **16** to device head **12**. The technique may include causing device head **12** to translate using the portion of the force transmitted through lead **16**. In some examples, the technique includes causing medical system **10** to translate by exerting a

distal or proximal force on device head **12** (e.g., on head first side **11**). The technique may include transmitting some portion of the force on device head **12** through lead **16** to IMD **14**. The technique may include causing IMD **14** to translate using the portion of the force transmitted through lead **16**. The technique may include using stylet **38** to impart a force on IMD **14** and/or device head **12**.

[0083] In examples, the technique includes translating device head **12** and at least some portion of lead **16** through a portion of lumen **20** within catheter distal section **24**. The technique may include translating device head **12** toward a lumen opening **28** at catheter distal end **30** of delivery catheter **18**. In examples, the technique include using delivery catheter **18** to substantially resist a rotation of device head **12** as device head **12** translates toward lumen opening **28**. The technique may include using delivery catheter **18** to cause a head second side **13** to remain distal to a head first side **11** opposite head second side **13** as device head **12** translates toward lumen opening **28**.

[0084] The technique may include translating a fixation element **21** coupled to device head **12** distally through lumen opening **28** as device head **12** translates toward lumen opening **28**. The technique may include causing fixation element **21** to engage tissues within target site **32** by translating fixation element **21** through lumen opening **28**. In examples, the technique includes extending fixation element **21** distally from head second side **13**. In examples, the technique includes causing a free end (e.g., free end **148** (FIG. 4) of a fixation tine **23** to penetrate tissues within target site **32**. The technique may include causing fixation tine **23** to expand radially outward as fixation element **21** translates through lumen opening **28**.

[0085] The technique may include causing electrode **19** on device head **12** (e.g., on head second side **13**) to contact tissues within target site **32** by translating device head **12** through lumen opening **28**. The technique may include causing electrode **19** to contact the tissues using fixation element **21**. The technique may include substantially securing device head **12** within the atrium of heart **26** using fixation element **21**. The technique may include substantially securing device head **12** within the atrium (e.g., the RA) of heart **26** while IMD **14** is positioned within the vena cava (e.g., the IVC) of heart **26**.

[0086] The technique may include causing medical system **10** to translate through lumen opening **28** by proximally withdrawing delivery catheter **18**. The technique may include substantially retaining device head **12** within the atrium using fixation element **21** and substantially retaining IMD **14** within the vena cava by proximally withdrawing delivery catheter **18**.

[0087] The disclosure includes the following examples.

[0088] Example 1: A medical system comprising: an implantable medical device having a proximal end and a distal end, wherein the implantable medical device is configured to be positioned within a vena cava of a heart; a flexible lead, wherein a proximal end of the flexible lead is attached to the distal end of the implantable medical device, wherein the flexible lead is configured to extend from the vena cava to an atrium of a heart; a device head, wherein a first side of the device head is attached to a distal end of the flexible lead, wherein the device head is configured to be positioned within the atrium of the heart; a fixation element attached to the device head, wherein the fixation element is configured to engage tissue to substantially secure the device head within the atrium of the heart; and an electrode coupled to the device head, wherein the electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the atrium of the heart.

[0089] Example 2: The medical system of example 1, wherein the electrode is electrically connected to circuitry within the implantable medical device.

[0090] Example 3: The medical system of example 1 or example 2, wherein the electrode is coupled to a second side of the device head opposite the first side.

[0091] Example 4: The medical system of any of examples 1-3, wherein the fixation element comprises one or more fixation tines attached to the device head, wherein the one or more fixation tines are configured to extend distally from the device head.

[0092] Example 5: The medical system of any of examples 1-4, wherein the flexible lead defines

an axis from the proximal end of the flexible lead to the distal end of the flexible lead, and wherein the flexible lead is configured to transmit a force along the axis from the implantable medical device or the device head to the other of the implantable medical device or the device head.

[0093] Example 6: The medical system of example 5, wherein the flexible lead is configured to have an axial stiffness relative to the axis greater than a bending stiffness relative to the axis.

[0094] Example 7: The medical system of any of examples 1-6, wherein the electrode is configured to engage a wall of the atrium via non-invasive physical contact when the device is positioned within the atrium.

[0095] Example 8: The medical system of any of examples 1-7, further comprising an electrode support member, wherein a proximal end of the electrode support member is attached to the second side of the device head and the electrode is located at a distal end of the electrode support member.

[0096] Example 9: The medical system of example 8, wherein the electrode support member is configured to advance the electrode into a tissue wall of the atrium by piercing the tissue wall of the atrium when the device is positioned within the atrium.

[0097] Example 10: The medical system of example 9, wherein the electrode support member is configured to advance the electrode into ventricular tissue of the heart.

[0098] Example 11: The medical system of any of examples 1-10, further comprising a stylet having a proximal end and a distal end, wherein the stylet is configured to impart an axial force in a distal direction to the device head, and wherein the first side of the device head is configured to receive the axial force from the distal end of the stylet.

[0099] Example 12: The medical system of any of examples 1-11, further comprising a device fixation element coupled to the implantable medical device, wherein the device fixation element is configured to position the implantable medical device within the vena cava of the heart.

[0100] Example 13: The medical system of any of examples 1-12, wherein the fixation member includes one or more fixation tines, wherein an individual fixation tine in the one or more fixation tines comprises a fixed end attached to the device head and a free end opposite the fixed end, and wherein the individual fixation tine is resiliently biased to pivot the free end radially outward from a longitudinal axis defined by the device head.

[0101] Example 14: The medical system of example 13, wherein the individual fixation tine includes a midpoint between the fixed end and the free end, wherein the individual fixation tine is resiliently biased to establish a configuration wherein the midpoint is distal to the free end and distal to the fixed end.

[0102] Example 15: The medical system of any of examples 1-14, wherein the fixation member includes a plurality of fixation tines, wherein each fixation tine in the plurality of fixation tines comprises a fixed end attached to the device head and a free end opposite the fixed end, such that the plurality of fixation tines comprise a plurality of fixed ends, and wherein the plurality of fixed ends surround the electrode.

[0103] Example 16: The medical system of any of examples 1-15, further comprising a delivery catheter defining a lumen, wherein: the lumen is configured to surround at least the implantable medical device, the flexible lead, the device head, and the fixation element; the lumen defines a lumen axis; and the implantable medical device, the flexible lead, the device head, and the fixation element are slidable along the lumen axis.

[0104] Example 17: The medical system of example 16, wherein the delivery catheter includes a proximal section and a distal section, wherein the distal portion includes a shape-memory alloy configured to define a curvature.

[0105] Example 18: The medical system of example 17, wherein the proximal portion is configured to at least partially reside within the vena cava when the distal section at least partially resides in the atrium.

[0106] Example 19: The medical system of example 17 or example 18, wherein the shape-memory alloy is configured to define a first curvature and a second curvature, wherein the second curvature

is out-of-plane with the first curvature.

[0107] Example 20: The medical system of any of examples 17-19, wherein the proximal portion is configured to surround the implantable medical device when the distal portion surrounds the device head and the fixation element.

[0108] Example 21: The medical system of any of examples 17-20, wherein the delivery catheter defines a lumen opening at a distal end of the delivery catheter, and wherein the fixation element, the device head, the flexible lead, and the implantable medical device are configured to pass therethrough.

[0109] Example 22: The medical system of example 22, wherein the fixation element comprises one or more fixation tines, wherein the one or more fixation tines are configured to expand radially outward when the delivery catheter is proximally withdrawn from a first position wherein the one or more fixation tines are proximal to the lumen opening to second position wherein the one or more fixation tines are distal to the lumen opening.

[0110] Example 23: The medical system of any of examples 1-22, wherein the flexible lead is configured to position the implantable medical device within the vena cava of the heart when the fixation element substantially secures the device head within the atrium of the heart.

[0111] Example 24: A system comprising: a medical assembly comprising: an implantable medical device having a proximal end and a distal end, wherein the implantable medical device is configured to be positioned within a vena cava of a heart; a flexible lead, wherein a proximal end of the flexible lead is attached to the distal end of the implantable medical device, wherein the flexible lead is configured to extend from the vena cava to an atrium of a heart; a device head, wherein a first side of the device head is attached to a distal end of the flexible lead, wherein the device head is configured to be positioned within the atrium of the heart; a fixation element attached to the device head, wherein the fixation element is configured to engage tissue to substantially secure the device head within the atrium of the heart; and an electrode coupled to the device head, wherein the electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the atrium of the heart; and a delivery catheter defining a lumen and defining a lumen opening at a distal end of the delivery catheter, wherein: the lumen is configured to surround at least the implantable medical device, the flexible lead, the device head, and the fixation element, and the fixation element, the device head, the flexible lead, and the implantable medical device are configured to pass distally through the lumen opening.

[0112] Example 25: The medical system of example 24, wherein the flexible lead defines an axis from the proximal end of the flexible lead to the distal end of the flexible lead, and wherein the flexible lead is configured to transmit a force along the axis from the implantable medical device or the device head to the other of the implantable medical device or the device head.

[0113] Example 26: The medical system of example 24 or example 25, wherein the fixation element comprises one or more fixation tines attached to the device head, wherein the one or more fixation tines are configured to extend distally from the device head.

[0114] Example 27: The medical system of any of examples 24-26, wherein the delivery catheter includes a proximal portion and a distal portion, wherein the distal portion includes a shape-memory alloy configured to define a curvature.

[0115] Example 28: The medical system of example 27, wherein the delivery catheter is configured to reduce the curvature when the delivery catheter is proximally withdrawn to cause one or more portions of the medical assembly to pass through the lumen opening.

[0116] Example 29: The medical system of example 28, wherein the proximal portion is configured to at least partially reside within the vena cava when the distal section at least partially resides in the atrium.

[0117] Example 30: The medical system of any of examples 27-29, wherein the shape-memory alloy is configured to define a first curvature and a second curvature, wherein the second curvature is out-of-plane with the first curvature.

[0118] Example 31: A method comprising: positioning a device head, an electrode attached to the device head, and a fixation element attached to the device head within an atrium of a heart by transmitting an axial force to the device head; causing a flexible lead attached to the device head and attached to an implantable medical device to extend from the atrium of the heart to a vena cava of the heart using the transmitted axial force; and positioning the implantable medical device within the vena cava of the heart using the transmitted axial force.

[0119] Example 32: The method of example 31, further comprising translating the device head, the electrode, the fixation element, the flexible lead, and the implantable medical device through a delivery catheter.

[0120] Example 33: The method of examples 31 or example 32, further comprising positioning a proximal portion of the delivery catheter in the vena cava and positioning a distal portion of the delivery catheter in the atrium.

[0121] Example 34: The method of example 33, further comprising defining a curvature in the distal portion of the delivery catheter when the distal portion of the catheter is positioned in the atrium.

[0122] Example 35: The method of example 34, further comprising causing the delivery catheter to reduce the curvature by proximally withdrawing the delivery catheter to cause at least one of the device head, the lead, or the implantable medical device to pass through a lumen opening of the delivery catheter.

[0123] Example 36: The method of any of examples 31-35, further comprising exerting the axial force on the implantable medical device and transmitting some portion of the axial force to the device head using the flexible lead.

[0124] Example 37: The method of any of examples 31-36, further comprising exerting the axial force on the device head using an elongated member and transmitting some portion of the axial force to the implantable medical device using the flexible lead.

[0125] Example 38: The method of any of examples 31-37, further comprising causing the fixation element to engage a wall of the atrium when the device head, the electrode, and the fixation element attached are positioned within the atrium.

[0126] Various examples of the disclosure have been described. Any combination of the described systems, operations, or functions is contemplated. These and other examples are within the scope of the following claims.

Claims

1. A medical system comprising: an implantable medical device having a proximal end and a distal end, wherein the implantable medical device is configured to be positioned in proximity to a heart; a flexible lead, wherein a proximal end of the flexible lead is attached to the distal end of the implantable medical device, wherein the flexible lead defines an axis from the proximal end of the flexible lead to a distal end of the flexible lead, and wherein the flexible lead is configured to have an axial stiffness relative to the axis greater than a bending stiffness relative to the axis; a device head, wherein a first side of the device head is attached to a distal end of the flexible lead, wherein the device head is configured to be positioned within a chamber of the heart, and wherein the flexible lead is configured to transmit a force along the axis from one of the implantable medical device or the device head to the other of the implantable medical device or the device head; a fixation element attached to the device head, wherein the fixation element is configured to engage tissue to substantially secure the device head within the chamber of the heart; a shallow electrode coupled to the device head, wherein the shallow electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the chamber of the heart; and a deep electrode coupled to the device head, wherein the deep electrode is configured to advance into the tissues of the heart and provide stimulation to the tissues of the heart when the device head is

positioned within the chamber of the heart.

2. The medical system of claim 1, wherein at least one of the shallow electrode or the deep electrode is electrically connected to circuitry within the implantable medical device.

3. The medical system of claim 1, wherein the shallow electrode is coupled to a second side of the device head opposite the first side.

4. The medical system of claim 1, wherein the fixation element comprises one or more fixation tines attached to the device head, wherein the one or more fixation tines are configured to extend distally from the device head.

5. The medical system of claim 1, wherein the implantable medical device is configured to be positioned with a vena cava of the heart, and wherein the flexible lead is configured to extend from the vena cava to the chamber of the heart.

6. The medical system of claim 1, wherein the deep electrode is coupled to a second side of the device head opposite the first side.

7. The medical system of any of claim 1, wherein the device head is configured to receive a distal end of a stylet, wherein the device head is configured to receive an axial force in a distal direction imparted by the stylet when the device head receives the distal end of the stylet.

8. The medical system of claim 1, wherein the flexible lead is configured to position the implantable medical device within a vena cava of the heart when the fixation element substantially secures the device head within the chamber of the heart.

9. The medical system of claim 1, wherein the fixation member includes one or more fixation tines, wherein an individual fixation tine in the one or more fixation tines comprises a fixed end attached to the device head and a free end opposite the fixed end.

10. The medical system of claim 9, wherein the individual fixation tine is resiliently biased to pivot the free end radially outward from a longitudinal axis defined by the device head.

11. The medical system of claim 1, further comprising a delivery catheter defining a lumen, wherein: the lumen is configured to surround at least the implantable medical device, the flexible lead, the device head, and the fixation element; the lumen defines a lumen axis; and the implantable medical device, the flexible lead, the device head, and the fixation element are slidable along the lumen axis.

12. The medical system of claim 11, wherein the delivery catheter includes a proximal section and a distal section, wherein the distal portion includes a shape-memory alloy configured to define a curvature.

13. The medical system of claim 12, wherein the shape-memory alloy is configured to define a first curvature and a second curvature, wherein the second curvature is out-of-plane with the first curvature.

14. The medical system of claim 11, wherein the delivery catheter defines a lumen opening at a distal end of the delivery catheter, and wherein the fixation element, the device head, the flexible lead, and the implantable medical device are configured to pass therethrough.

15. The medical system of claim 14, wherein the fixation element comprises one or more fixation tines, wherein the one or more fixation tines are configured to expand radially outward when the delivery catheter is proximally withdrawn from a first position wherein the one or more fixation tines are proximal to the lumen opening to a second position wherein the one or more fixation tines are distal to the lumen opening.

16. A system comprising: a medical assembly comprising: an implantable medical device having a proximal end and a distal end, wherein the implantable medical device is configured to be positioned in proximity to a heart, a flexible lead, wherein a proximal end of the flexible lead is attached to the distal end of the implantable medical device, wherein the flexible lead defines an axis from the proximal end of the flexible lead to a distal end of the flexible lead, and wherein the flexible lead is configured to have an axial stiffness relative to the axis greater than a bending stiffness relative to the axis, a device head, wherein a first side of the device head is attached to a

distal end of the flexible lead, wherein the device head is configured to be positioned within a chamber of the heart, and wherein the flexible lead is configured to transmit a force along the axis from one of the implantable medical device or the device head to the other of the implantable medical device or the device head, a fixation element attached to the device head, wherein the fixation element is configured to engage tissue to substantially secure the device head within the chamber of the heart, a shallow electrode coupled to the device head, wherein the shallow electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the atrium of the heart, and a deep electrode coupled to the device head, wherein the deep electrode is configured to advance into the tissues of the heart and provide stimulation to the tissues of the heart when the device head is positioned within the chamber of the heart; and a delivery catheter defining a lumen and defining a lumen opening at a distal end of the delivery catheter, wherein: the lumen is configured to surround at least the implantable medical device, the flexible lead, the device head, and the fixation element, and the fixation element, the device head, the flexible lead, and the implantable medical device are configured to pass distally through the lumen opening.

17. The medical system of claim 16, wherein the delivery catheter includes a proximal portion and a distal portion, wherein the distal portion includes a shape-memory alloy configured to define a first curvature and a second curvature, wherein the second curvature is out-of-plane with the first curvature.

18. The medical system of claim 16, wherein the flexible lead is configured to position the implantable medical device within a vena cava of the heart when the fixation element substantially secures the device head within the chamber of the heart.

19. A method comprising: positioning a device head, an shallow electrode attached to the device head, a deep electrode attached to the device head, and a fixation element attached to the device head within a chamber of a heart by transmitting an axial force to the device head, wherein a first side of the device head is attached to the a distal end of a flexible lead, wherein the electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the chamber of the heart, wherein the fixation element is configured to engage tissue to substantially secure the device head within the chamber of the heart, wherein the shallow electrode is configured to provide stimulation to tissues of the heart when the device head is positioned within the chamber of the heart, and wherein the deep electrode is configured to advance into the tissues of the heart and provide stimulation to the tissues of the heart when the device head is positioned within the chamber of the heart; and causing the flexible lead attached to the device head and attached to an implantable medical device to extend into the chamber of the heart using the transmitted axial force, wherein a proximal end of the flexible lead is attached to a distal end of the implantable medical device, wherein the flexible lead defines an axis from the proximal end of the flexible lead to the distal end of the flexible lead, wherein the flexible lead is configured to have an axial stiffness relative to the axis greater than a bending stiffness relative to the axis, and wherein the flexible lead is configured to transmit the axial force along the axis from one of the implantable medical device or the device head to the other of the implantable medical device or the device head.

20. The method of claim 19 further comprising translating the device head, the electrode, the fixation element, the flexible lead, and the implantable medical device through a delivery catheter.
